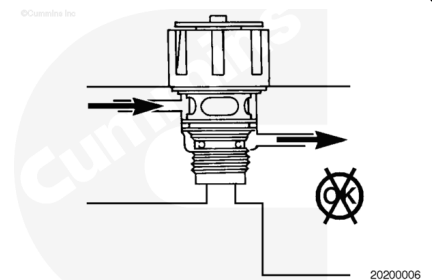


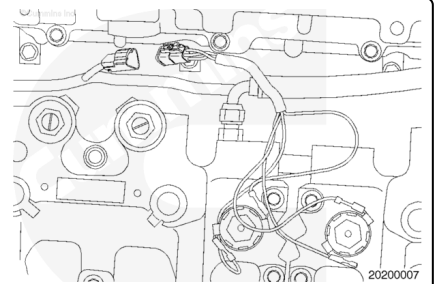
Trainings ▾

Initial Check

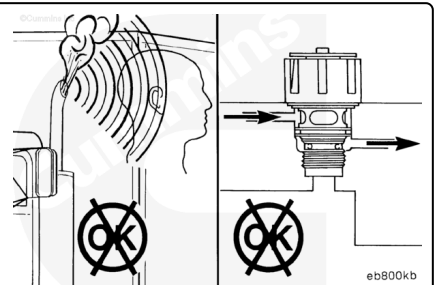
A stuck oil flow solenoid valve can prevent the engine brake from starting or performing properly.



To check for a stuck solenoid valve, disconnect the electrical connection to the engine brakes and operate the engine.

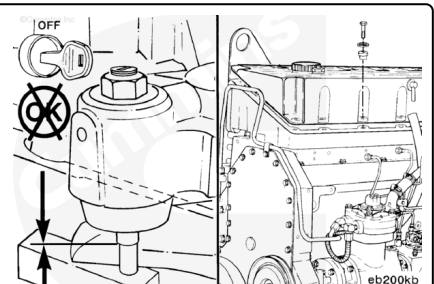


If exhaust gases can be heard escaping past the valves, the solenoid valve or valves **must** be replaced. Contact an authorized engine brake dealer.



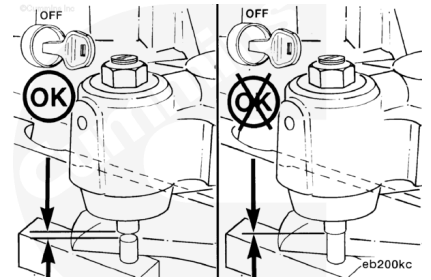
An incorrectly adjusted or damaged secondary piston will cause the engine to run rough or **not** to start.

To check the secondary piston, remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(</qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html>)



Check the secondary piston clearance.

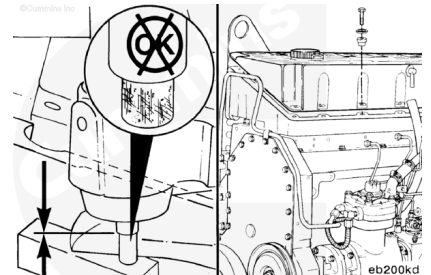
Refer to Engine Brake - Adjustment which appears later in this procedure to adjust the clearance.



Inspect the secondary piston for damage.

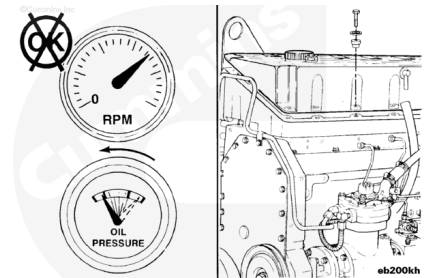
If the secondary piston is damaged, contact an authorized engine brake dealer for repair.

Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html>)



If a sudden drop in lubricating oil pressure causes the engine to **not** start or run rough, perform the following steps.

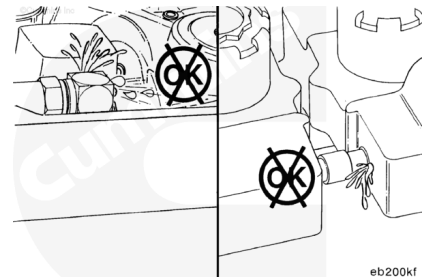
Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html>)



Operate the engine and check for the following problems:

- Oil supply hose loose or missing
- Oil connector seal loose or missing.

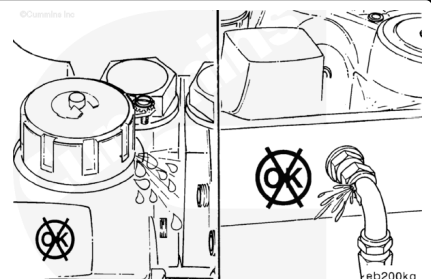
Contact an authorized engine brake dealer to correct the problem.



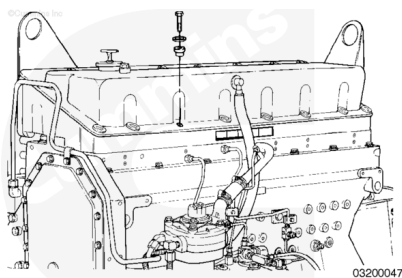
If the oil supply hose or the connector seal was **not** the problem, check for the following:

- Upper solenoid valve seal ring missing or damaged
- External oil supply hose or fitting cracked or leaking.

Contact an authorized engine brake dealer to correct the problem.

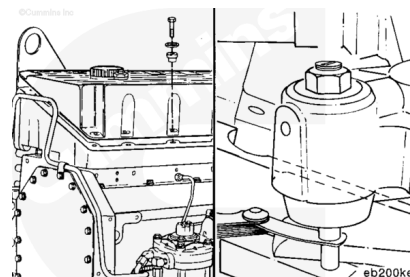


Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html>)

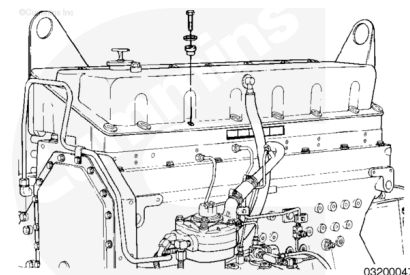


If the engine is misfiring or has a loss of power:

- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html>)
- Adjust the secondary piston clearance. See the Engine Brake - Adjustment which appears later in this procedure.

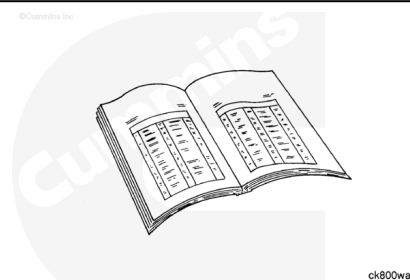


Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html>)



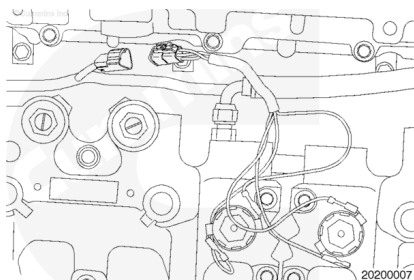
Preparatory Steps

- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html>)

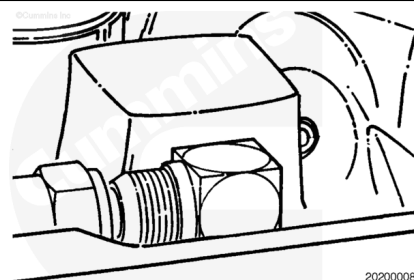


Remove

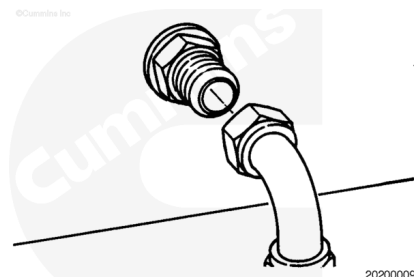
Disconnect the solenoid power and ground wires from the terminals on the inside of the rocker lever housing spacer.



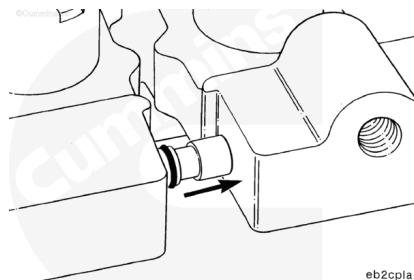
Disconnect the internal oil supply hose connection.



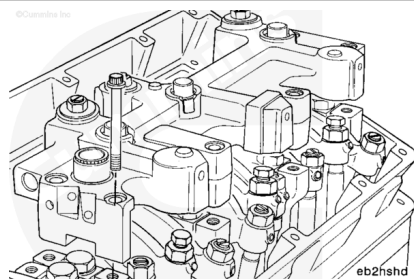
Disconnect the external oil supply hose connection.



Press the connector into the front housing to allow clearance for housing removal.



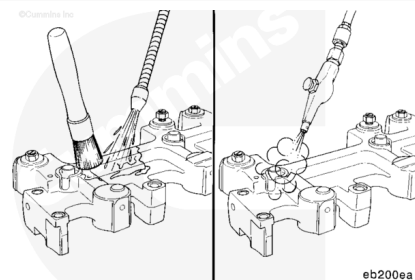
Remove the capscrews and both engine brake housings.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

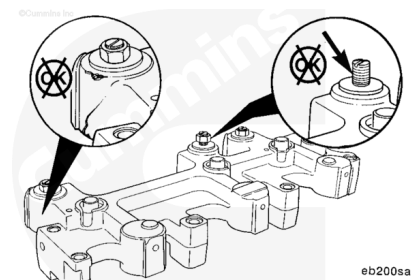


⚠ WARNING ⚠

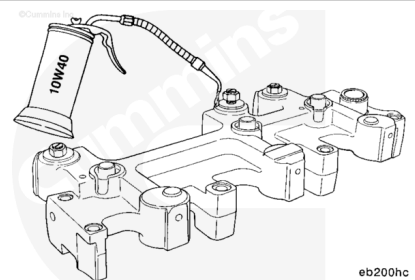
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use an approved solvent to clean the brake housings. Dry with compressed air.

Inspect the housings for any missing or damaged parts.



Apply a thin film of clean 15W-40 oil to the machined surfaces.

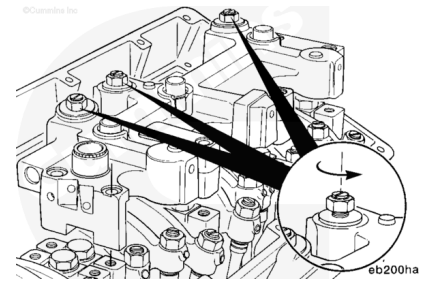


Install

Loosen the locknuts on the secondary pistons.

Make sure the secondary pistons are fully retracted.

Install the rear engine brake housing on the rear rocker lever supports.

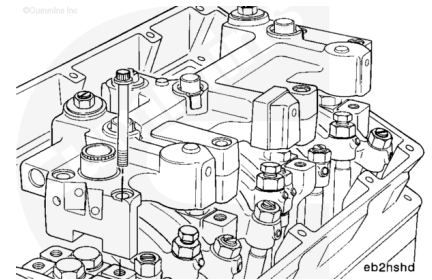


Use clean 15W-40 oil to coat the threads and the bottom side of the capscrew head.

Note : Washers are not used with the capscrews.

Install the capscrews in the rocker lever supports.

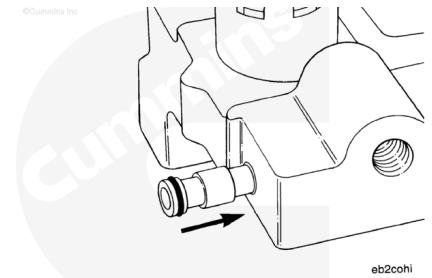
Do **not** tighten the capscrews.



Use clean 15W-40 oil to lubricate the o-rings.

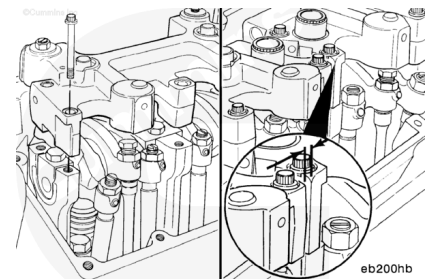
Press the connector all the way into the front housing by hand.

When installing the front housing, be sure the oil connector and o-ring are in a position to be pushed into the rear housing.



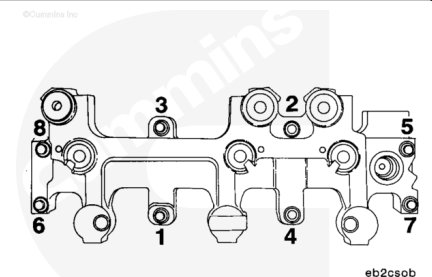
Install the front engine brake housing on the front rocker lever supports.

Center the oil connector between the front and rear housings before tightening the capscrews.



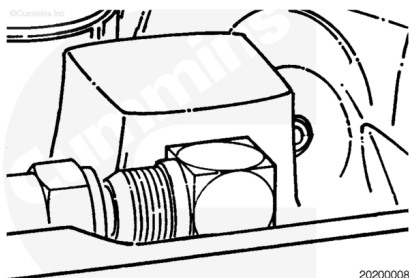
Tighten the capscrews in the sequence shown.

Torque Value: 81 n•m [60 ft-lb]



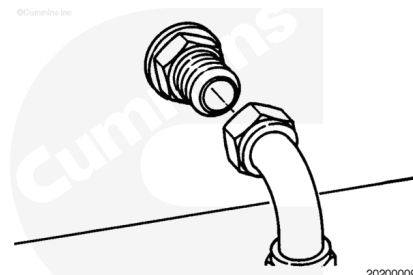
Install the internal oil connector supply hose and tighten the capscrews.

Torque Value: 19 n•m [168 in-lb]

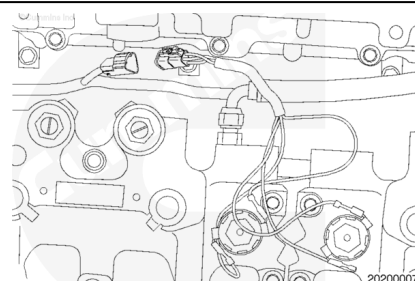


Install the external oil connector supply hose and tighten the capscrews.

Torque Value: 19 n•m [168 in-lb]



Connect the solenoid power and ground wires to the terminals on the inside of the rocker lever housing, red wire to the rear housing solenoid and white wire to the front housing solenoid.



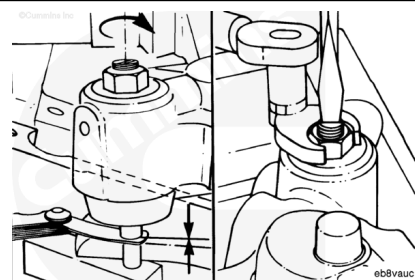
Adjust

⚠ CAUTION ⚠

Do not tighten the adjusting screw too tightly. The engine can be damaged.

⚠ CAUTION ⚠

Failure to set the correct lash can damage the engine.



To obtain maximum brake operating efficiency, and prevent engine damage by piston-to-valve contact, complete the following instructions carefully.

Adjust the exhaust valves on the appropriate cylinder. Refer to Procedure 003-004 in Section 3.

(/qs3/pubsys2/xml/en/procedures/35/35-003-004-tr.html)

After adjusting the exhaust valves on the appropriate cylinder, insert the correct feeler gauge.

Note : Engine brake slave piston lash varies depending on the brake model fitted and may differ between front and rear brake assemblies on the same engine. The correct slave piston lash must be obtained from the dataplate attached to the engine brake housing.

Hold the screw in position and tighten the locknut.

Torque Value:

Without the Torque Wrench Adapter

1.34 n•m [25 ft-lb]

Torque Value:

With the Torque Wrench Adapter Part Number 3163196

1.30 n•m [22 ft-lb]

After the secondary piston adjusting screw locknut is tightened to the correct torque value, check the clearance with the feeler gauge again.

Finishing Steps

- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html>)

©Cummins Inc.



ck800wa